DM-EXton2 Teleoperation Data Acquisition System Product User Manual V1.0

1 Product Overview

2 Before Use

3 Wear and Connect

4 Instructions for Using Software and Hardware

5 Safety and Precautions

6 Maintenance and Upkeep

7 Disclaimer

8 Contact Information

1 Product Overview

1.1 Product Introduction

DM-EXton2 is a new generation of wearable teleoperated data acquisition device launched by Daimon. Centering on high precision and low latency, it combines the characteristics of portability, efficiency, and safety, and is committed to providing users with professional and reliable data acquisition solutions to break the deadlock in data acquisition in the embodied intelligence industry. The device can be used to teleoperate current mainstream dual-arm robots, completing fine and complex operations from simple grasping and releasing, tool use, to folding clothes and storing items, providing support for robots to stably and efficiently acquire high-quality multimodal data.

Building on the core advantages of its predecessor, such as low latency, high precision, and excellent heterogeneous compatibility, DM-EXton2 has optimized its overall structure, making it more suitable for the physical needs of most remote operation data acquisition personnel and enhancing operational comfort. Meanwhile, the modular design supports dual-end adaptation of both control pads and gloves, improving replacement efficiency and multi-scenario adaptability, and further upgrading operational convenience. The addition of whole-body remote operation control, three-level motion mapping proportional scaling, and interactive force feedback functions not only enables flexible operation but also effectively addresses the

operational deviation issue caused by the "lack of feedback" in traditional remote operation, making data acquisition easier and more efficient.

1.2 Product Features

1.2.1 DM-EXton2 P Teleoperation Data Acquisition System Backpack Version

- **Fine Force/Tactile Feedback**

Integrated self-developed visual-tactile sensor, accurately restoring light touch sensations

- **1000Hz Low Latency**

Millisecond-level instruction synchronization and response, with seamless human-machine collaboration without lag

- **Full-body remote operation support**

Supports whole-body coordinated teleoperation of the robot (including the mobile chassis and waist joints), covering a wider range of Data Acquisition operations

- **Three-level motion proportional scaling**

One-click switch between master-slave mapping relationships, catering to both fine operation and large-scale motion scenarios

- **Multi-modal End Adaptation**

The snap-on quick-change design allows for rapid switching between control pads and gloves, flexibly adapting to various end-effectors, achieving "one machine with multiple uses"

1.2.2 DM-EXton2 G Controller

- Supports full-body teleoperation
- Instantaneous response, no time delay, output frequency 1000Hz
- Ergonomic Design

1.3 Product Specification Parameters

1.3.1 DM-EXton2 P Teleoperation Data Acquisition System Backpack Version

Table 1-1 DM-EXton2 P Equipment Specification Parameter Table

Product Weight	4.0kg
Product Dimensions	589mmx200mmx808mm
Sampling Frequency	Up to 1000Hz

Communication Method	Ethernet/WiFi
API Interface	C++\ROS 1\ROS 2
Encoder Type	Single-turn absolute encoder
Encoder Angular Resolution	≤0.01 °
Battery Specifications	25.9V 5Ah
Endurance Time	Approximately 6h
Charging Time	≤2h
Light Indication	Indicators for ontology status, fault alarm, etc.
Warranty Period	1 year
Operating Temperature	0°C~40°C
Operating Humidity	10~90%RH without condensation
Air Environment	No dust, flammable, explosive, or corrosive gases

1.3.2 DM-EXton2 G **Control pad**

Table 1-2 DM-EXton2 G Specification Parameter Table

Product Weight (One Hand)	0.28kg
Product Dimensions	254mmx83mmx48mm
Encoder Type	Single-turn absolute encoder
Encoder angular resolution	≤0.01 °
Operating Temperature	0°C~40°C
Sampling Frequency	Up to 1000Hz
Warranty Period	1 year

1.4 Packing List

Table 1-3 DM-EXton2 P Teleoperation Data Acquisition System Backpack Edition Packaging List Table

Package Contents	Quantity	Unit
Dual-arm remote control device	1	set
Backpack Base (including battery)	1	pcs
Power Adapter	1	pcs
Calibration tool (cylindrical pin + fixed plastic part)	1	set
Inspection Certificate	1	pcs

Table 1-4 DM-EXton2 G Control pad Packaging List Table

Package Contents	Quantity	Unit
Control pad	1	set
Inspection Certificate	1	pcs

1.5 Dimensions

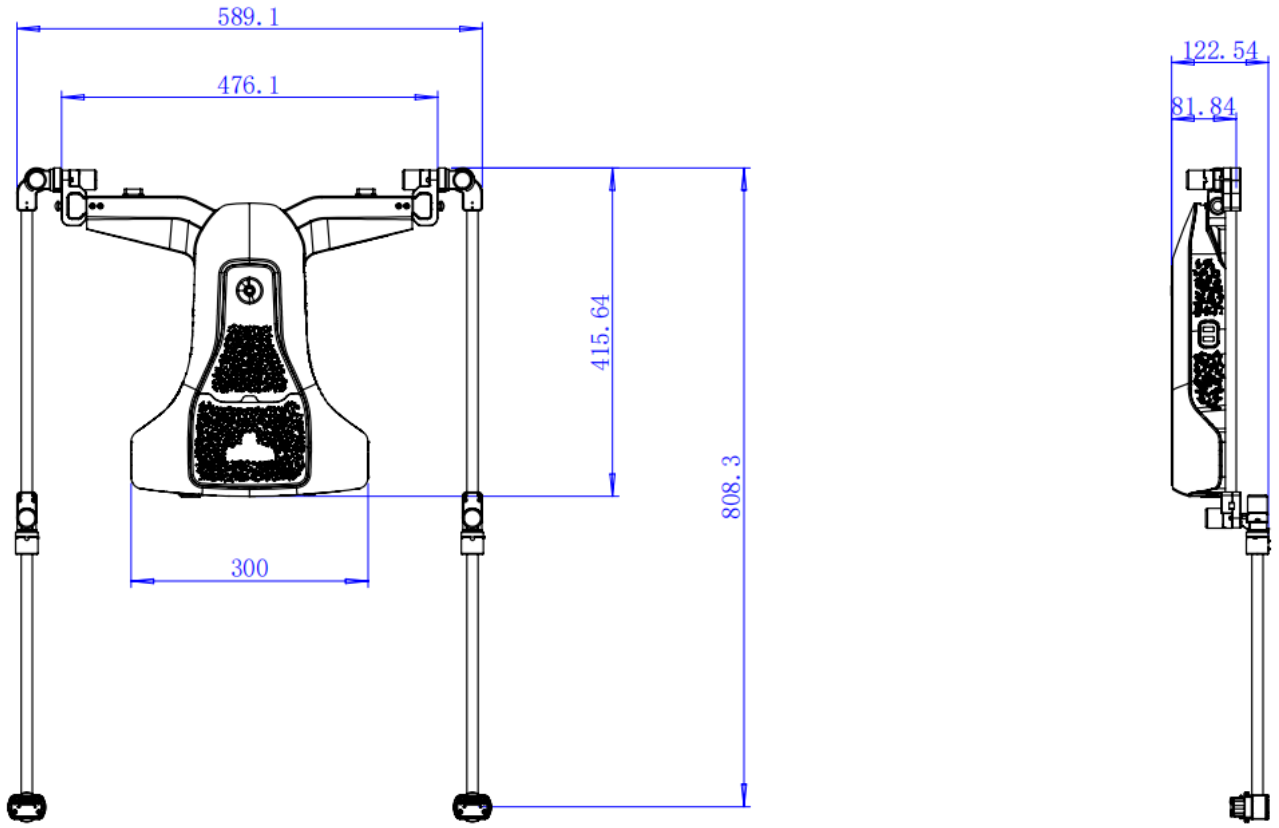


Figure 1-1 Dimensions of DM-EXton2 P Teleoperation Data Acquisition System, Unit: mm (excluding backpack fabric)

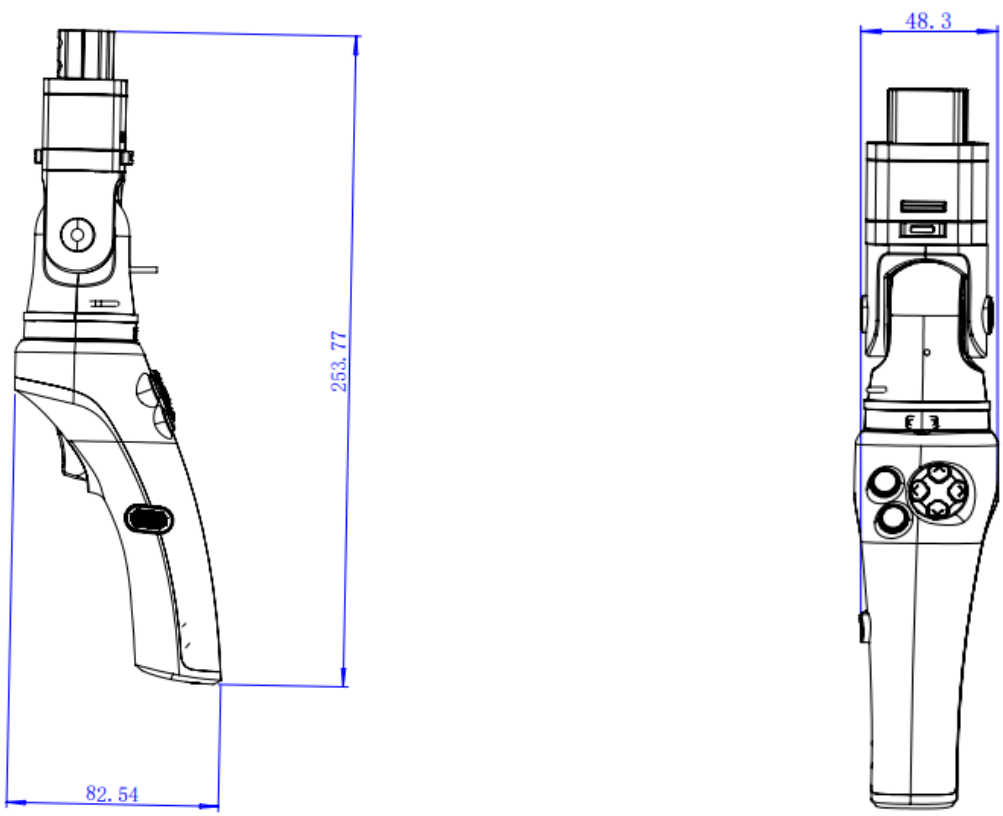


Figure 1-2 DM-EXton2 G Dimension Diagram, Unit: mm

2 Before Use

2.1 Unboxing Confirmation

Please perform unpacking confirmation according to 1.4 Packing List. The products and accessories inside the package are as shown in Figure 2-1 and Figure 2-2.



Figure 2-1 DM-EXton2 P Unboxing Diagram



Figure 2-2 DM-EXton2 G Unboxing Diagram

2.2 Assembly

2.2.1. First take out the backpack and place it on the table, then take out the arm;

When removing the arm, pay attention to distinguish between left and right (labeled with L). If removing the left arm, it should be installed on the left shoulder connector accordingly.

- After the structural component is connected to the card slot, as shown in the figure below

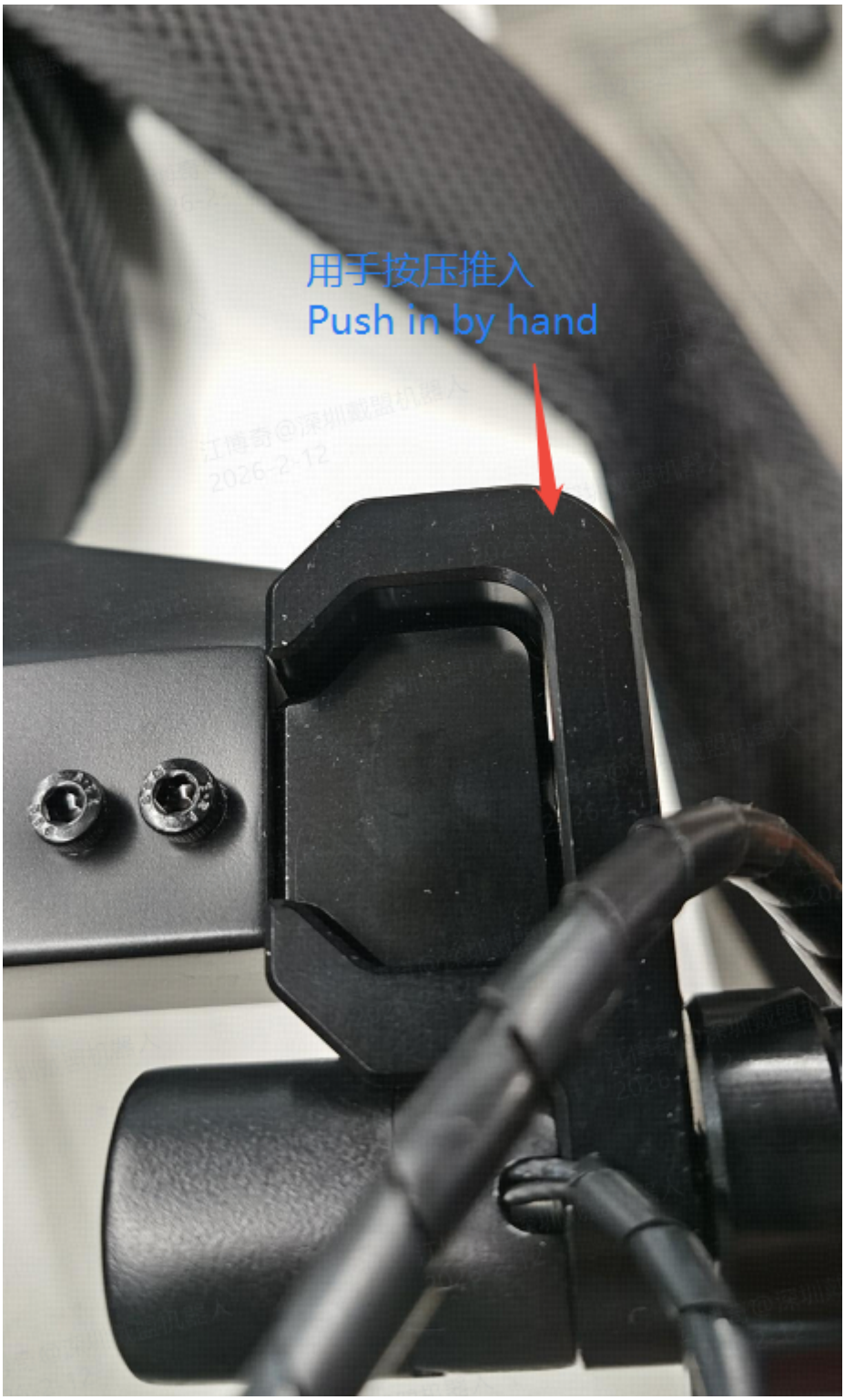


Figure 2-3 Structural Positioning Slots When the Arm Is Installed on the Backpack

- Insert and lock from the side using the buckle.



Figure 2-4 Structural Locking Buckle When the Arm is Installed on the Backpack

- Connect the cable connector

After straightening the connecting plug of the arm according to the posture marked in red in the figure below, insert it into the plug on the shoulder of the equipment backpack; the wiring harnesses must not be entangled or knotted together.

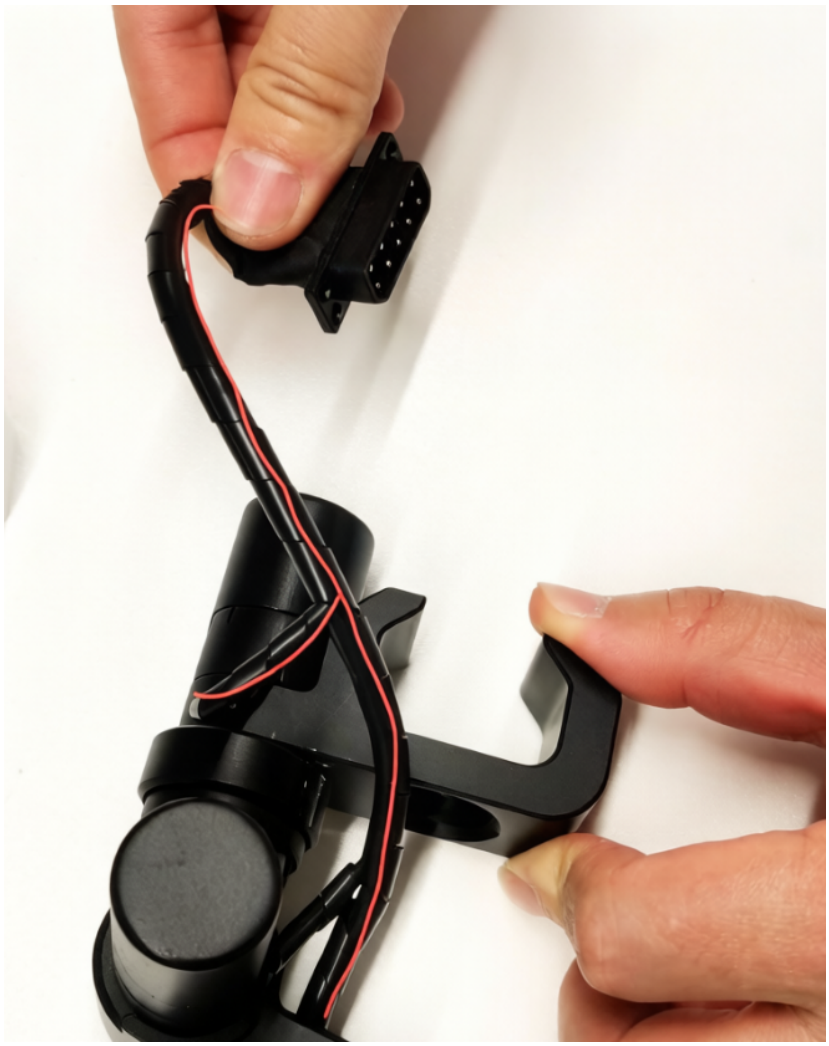


Figure 2-5 Data Cable Placement When the Arm Is Installed on the Backpack

- Overall view after arm installation is completed



Figure 2-6 Arm Installation Completed

- Arm Elbow Limit
  - After the arm is removed, check the elbow cable to ensure that the wiring harness is not wrapped around the shaft;

- Unscrew the limit screw as shown in the figure below by 2mm to limit the joint and prevent the wiring harness from getting entangled or even twisted off after the forearm rotates 360°;



Figure 2-7 Check the wiring harness and tighten the limit screw

- Control pad Installation



Figure 2-8 Schematic Diagram of Control pad Installation Alignment and Pressing

Installation of DM-EXton2 P and DM-EXton2 G ensures that the left arm uses the left Control pad, and the installation operation is to align and then press.

2.3 DM-EXton2 Buttons and Interfaces

As shown in Figure 2-9, the power on/off button of the teleoperation data acquisition device:



Figure 2-9 DM-EXton2 P Back Power On/Off Button and Indicator Lights

As shown in Figure 2-10, the Ethernet port and charging port of the remote operation data acquisition device:



Figure 2-10 DM-EXton2 P Bottom Ethernet Port and Charging Port

2.4 Power-on Confirmation

2.4.1 Power On/Off Button

- Power On: Press and hold the power button for 2 seconds to turn on the device. After powering on, the button indicator light will be white.**Stay on constantly**.

- Power Off: In the powered-on running state, long press the power on/off button for 3 seconds to turn off the device. After the device is turned off, the red indicator light on the button will go out.

2.4.2 Power On/Off Key Status

Power button indicator status:

- Normal operation: single flash, frequency 2Hz
- Work charging: Double flash
- Charging while powered off: three flashes
- Fill: Always On
- Low battery: Flash, frequency 4Hz

3 Wear and Connect

3.1 DM-EXton2 P Wear Recommendations

To ensure the consistency of movements between the wearer and the teleoperated object, as well as the comfort of the wearer when using this device, it is recommended that the wearer of this device:

Table 3-1 Recommendations for Wearers

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3.2 DM-EXton2 P Wear Steps

- Put on the shoulder strap of DM-EXton2 P and adjust it to a suitable length,
- Ensure that the shoulder parts of the dual arms of the teleoperation device are level with the human shoulders, and fasten the chest and waist straps.

As shown in Figure 3-1, schematic diagram of complete wearing



Figure 3-1 Schematic Diagram of Completed Wear

3.3 DM-EXton2 P 与 DM-EXton2 G 拆装

With one hand, use your fingers to press both ends of the spring key of the quick-release head, press it firmly, then with the other hand, grasp the other end of the quick-release and gently pull it out in coordination:



Figure 3-2 Schematic Diagram of Control pad Disassembly

Installation of DM-EXton2 P and DM-EXton2 G ensures that the left arm uses the left Control pad, and the installation operation is to align and then press.

4 Instructions for Using Software and Hardware

4.1 Hardware Usage Instructions

4.1.1 Description of the Dual-Arm Joints

When using the dual-arm teleoperation device, it is necessary to ensure that movements are carried out within the joint rotation angle range specified in Table 4-1. The markings of Joint 1 to Joint 6 are shown in Figure 4-1 (Note: Figure 4-1 is a diagram observed by an observer standing behind the device).

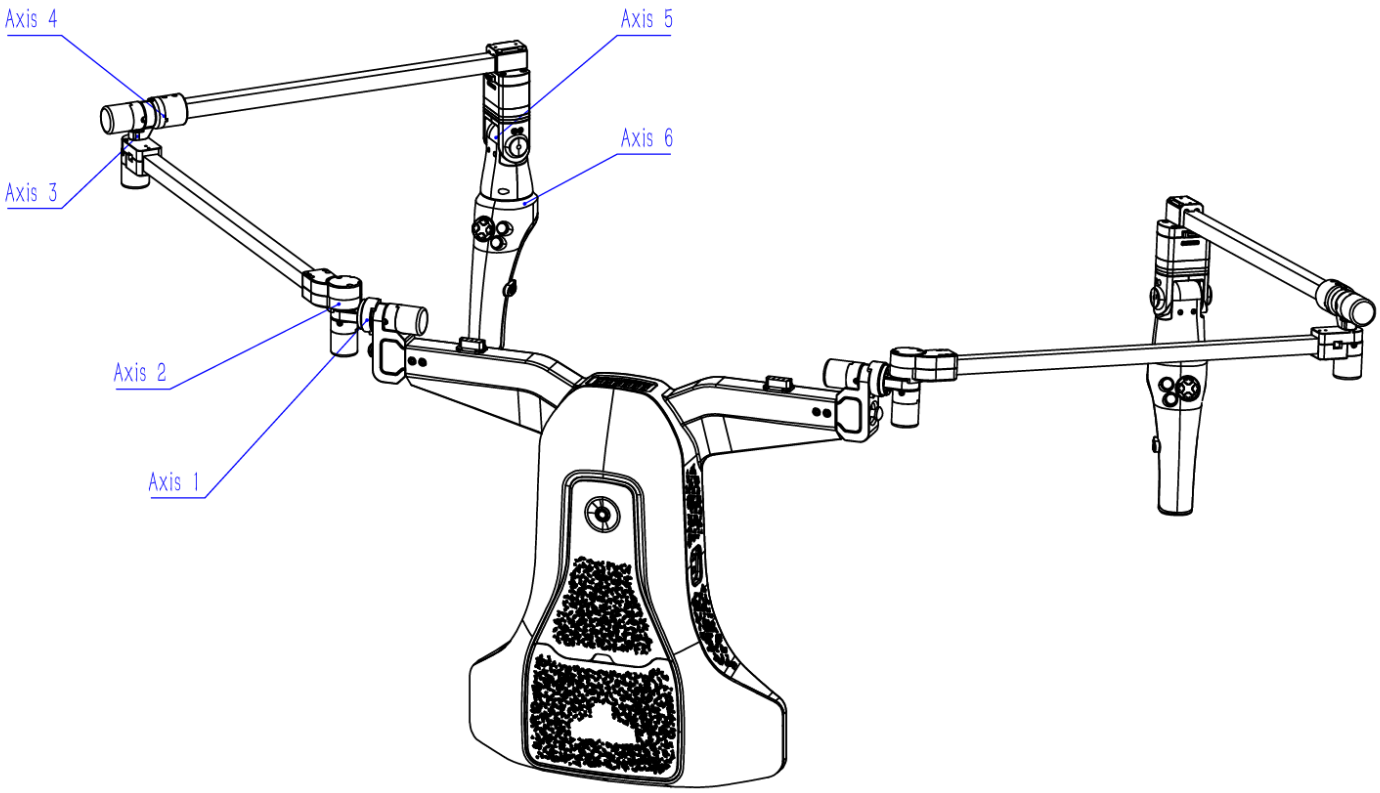


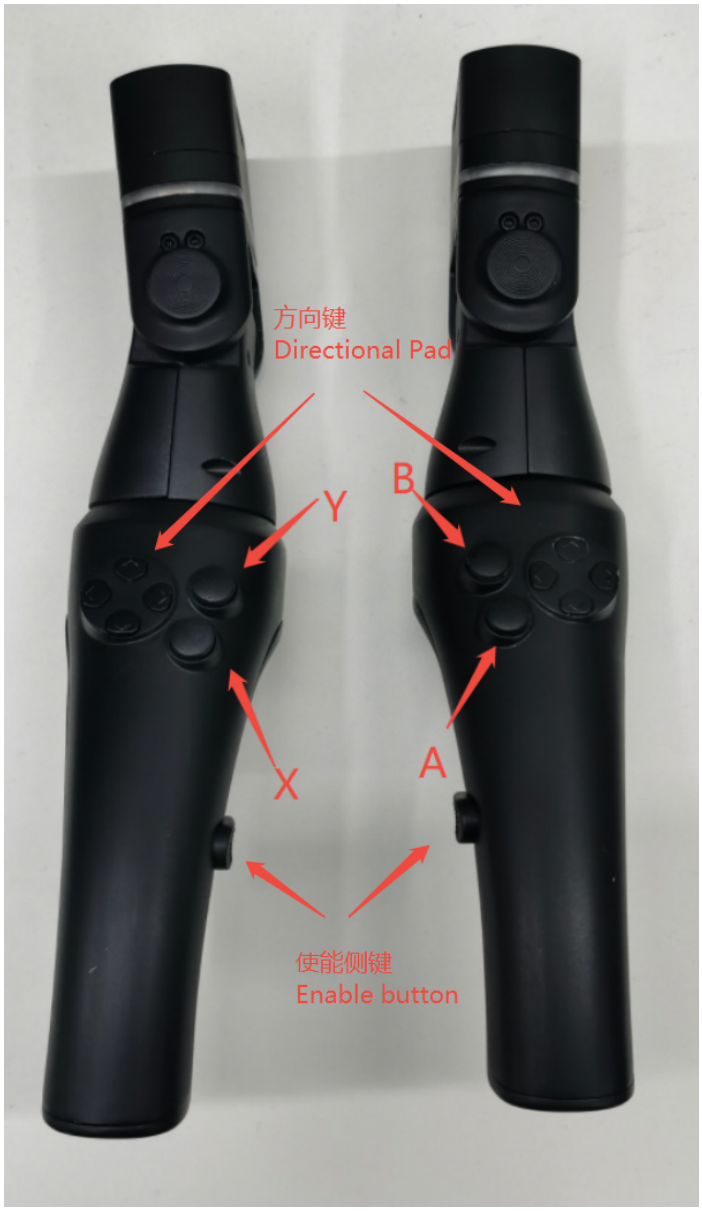
Figure 4-1 Joint Identification Diagram of Dual-Arm Teleoperation Device

Table 4-1 Joint Rotation Angle Range

关节旋转角度范围	
关节名称	旋转角度(相对于关节零位)
关节1	-5° 至+150°
关节2	-36° 至+90°
关节3	-23° 至+234°
关节4	-48° 至+228°
关节5	-7° 至+187°
关节6	-140° 至+180°

Mapping relationship between encoder IDs and each joint. As shown in Figure 4-1, each arm has 6 joints; the IDs corresponding to joints 1-6 in the figure are 0-5.

4.1.3 Instructions for Using the Control pad Device



When the connection is completed, the white light will illuminate; otherwise, the red light will illuminate.

The button definition can be set in conjunction with the API defined in 4.3.3;

The following is the recommended reference definition for the control pad:

Left controller:

Directional cross key: Controls the movement of the chassis or torso

Trigger: Controls the "opening and closing" of the robot's end effector.

Enable side key (grip key): When not pressed, motion control of the torso, arms, and chassis will not take effect.

X Key: Not yet defined.

Y key: Not yet defined.

Right controller:

Directional cross key: controls the movement of the chassis or torso

Trigger: Controls the "opening and closing" of the robot's end effector.

Enable side key (grip key): When not pressed, motion control of the torso, arms, and chassis will not take effect.

Button B: Click to start data acquisition, click again to save data and exit;

Key A: Data acquisition stops, data is neither saved nor written to disk;

4.1.4 Description of Network Self-Test at Startup

After the device is powered on, it will automatically start relevant detection. It checks the environment every 2 seconds and emits a beep sound.

Generally speaking, it is divided into the following three situations:

1. When the device is not connected to Wi-Fi and no wired network cable is plugged in, it cannot detect a network, which will trigger an infinite number of self-checks.
2. When the device is connected to a wired network cable and configured successfully as instructed below, it usually passes within 1-3 self-check cycles.
3. The device successfully configures Wi-Fi according to the instructions below and disconnects the Ethernet cable. The device usually passes within 1-5 self-check cycles.

4.1.5 Instructions for Using the Wired Network Port

The wired network port at the bottom of the device has a fixed IP:

```
代码块
1   192.168.2.240
```

When using a wired network, please connect the remote operation data acquisition device to the target machine via a network cable, and set the network port of the target device to:

```
代码块
1   192.168.2.xxx
```

Check if the connection is successful through the command:

```
代码块
1   ping 192.168.2.240
```

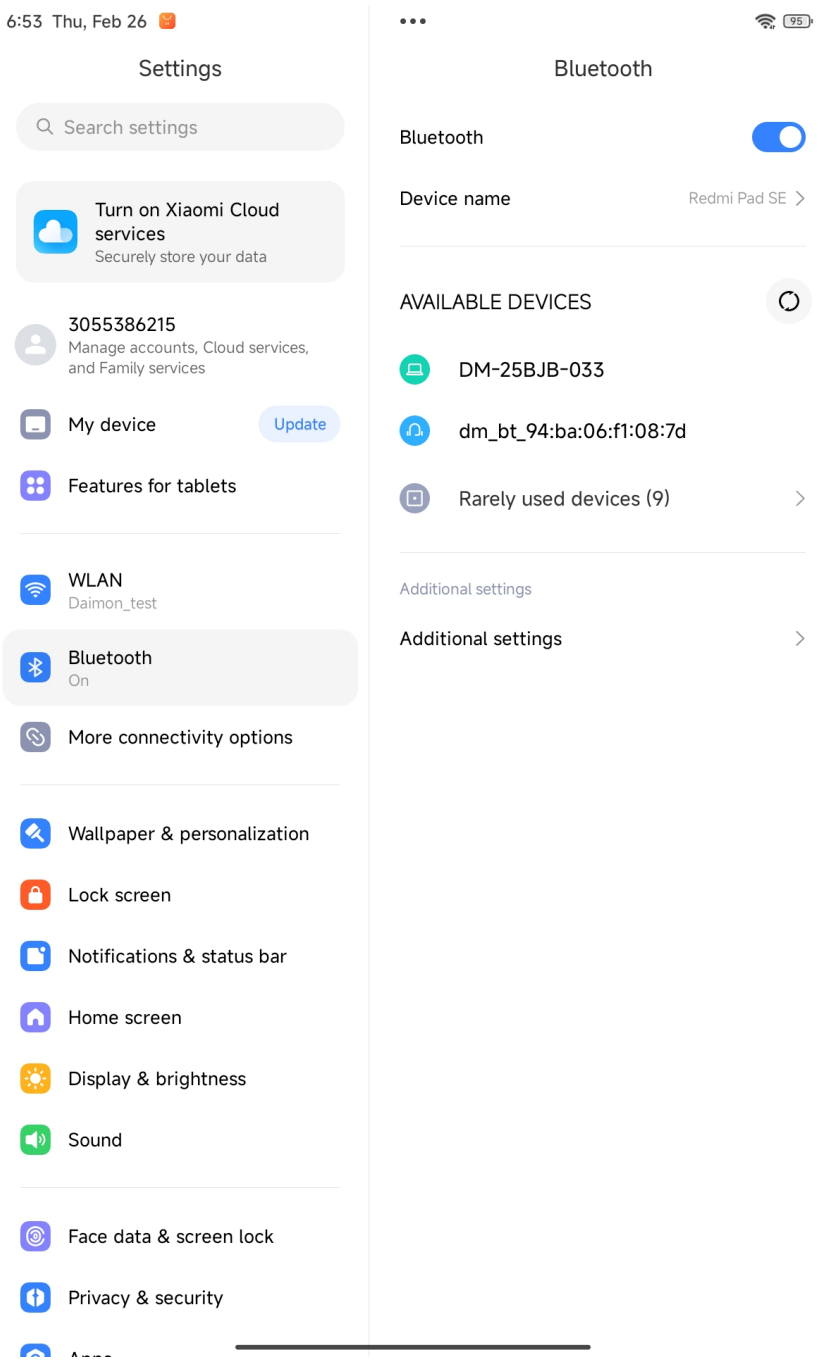
4.1.6 Instructions for Bluetooth Network Configuration of Wireless Network Card

Bluetooth network configuration requires installing the Android APK provided by us, installing it, and turning on the device.

Step 1:

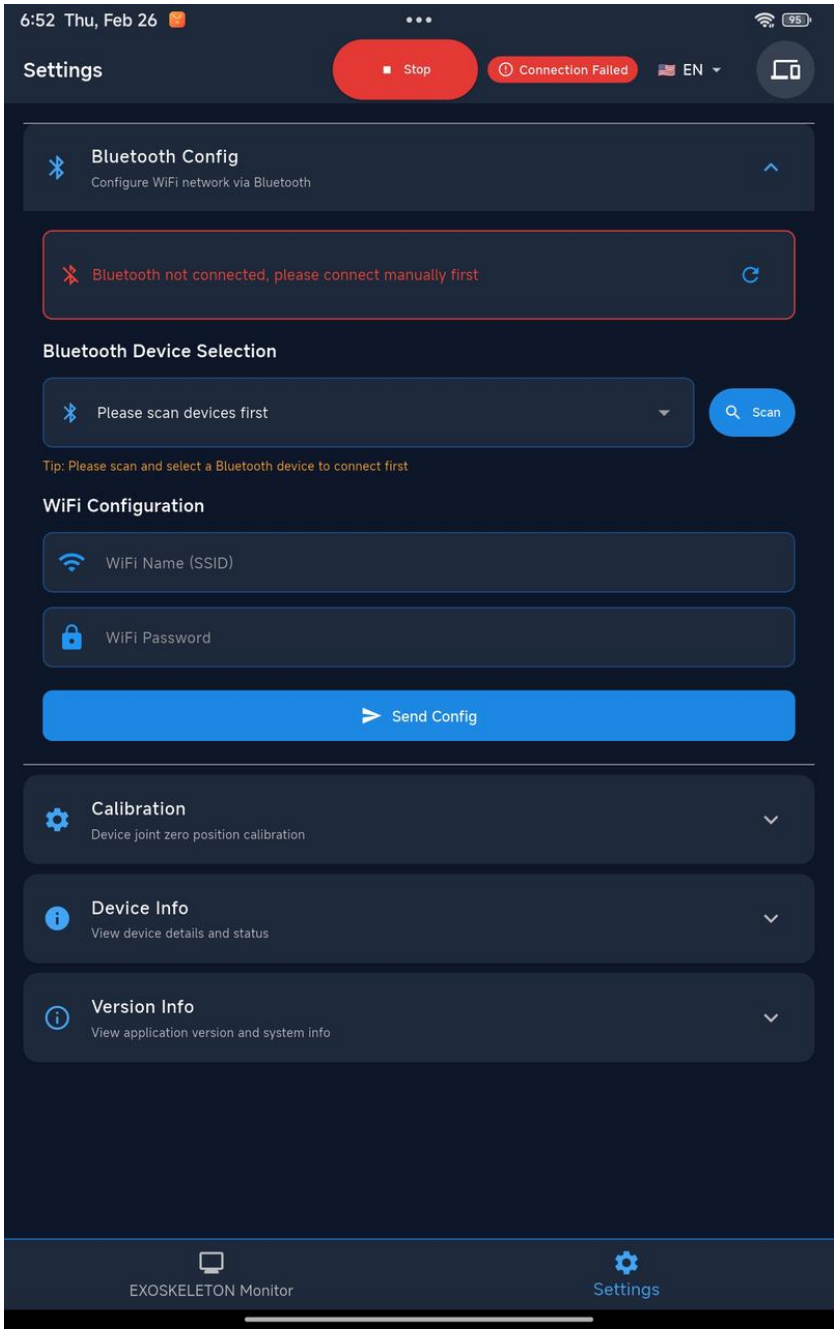
Turn on the Bluetooth of your phone and search for Bluetooth devices:

The Bluetooth name of the remotely operated data acquisition device is dm_bt_xxx, click to connect the device.



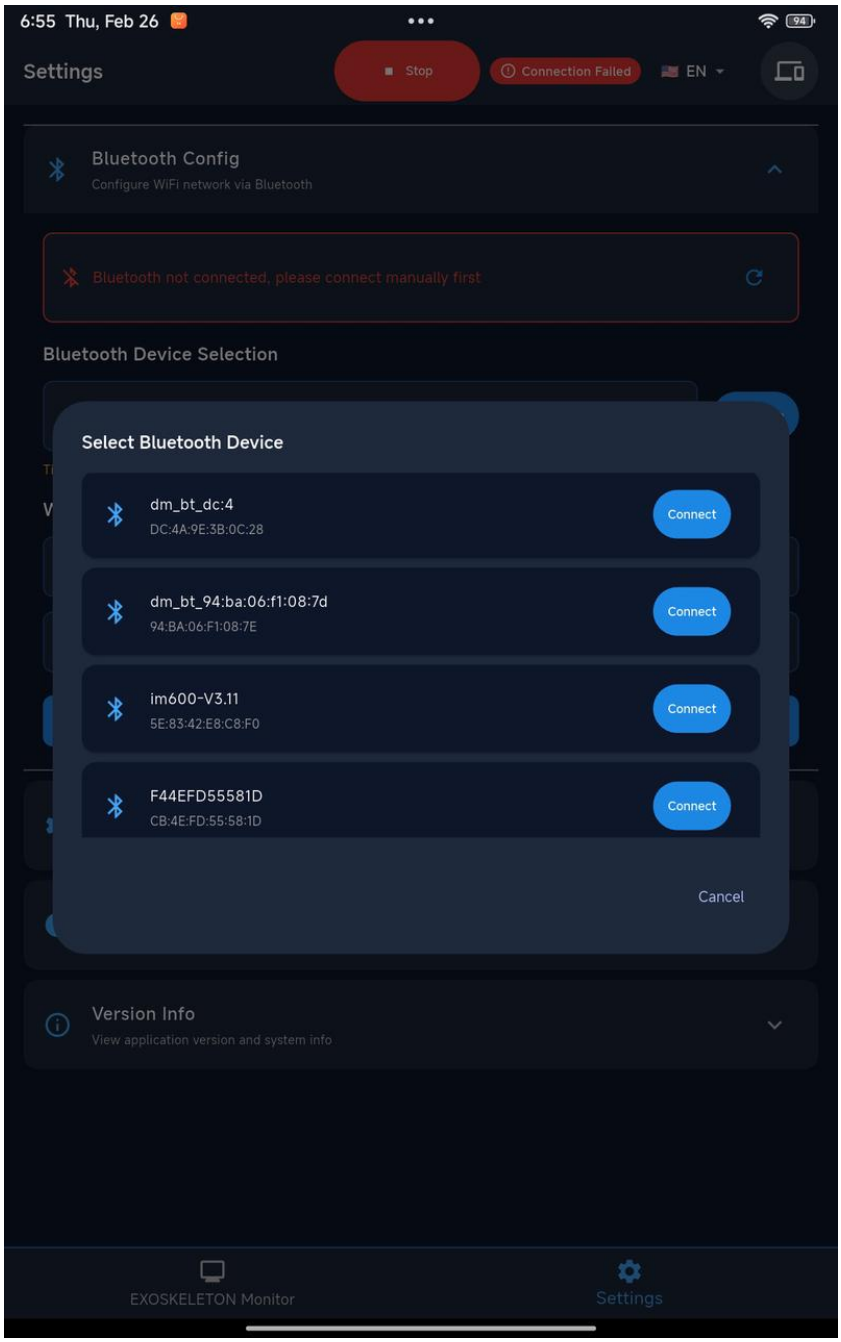
Step 2:

After the phone detects Bluetooth, open the App. Switch to Settings below.



Step 3:

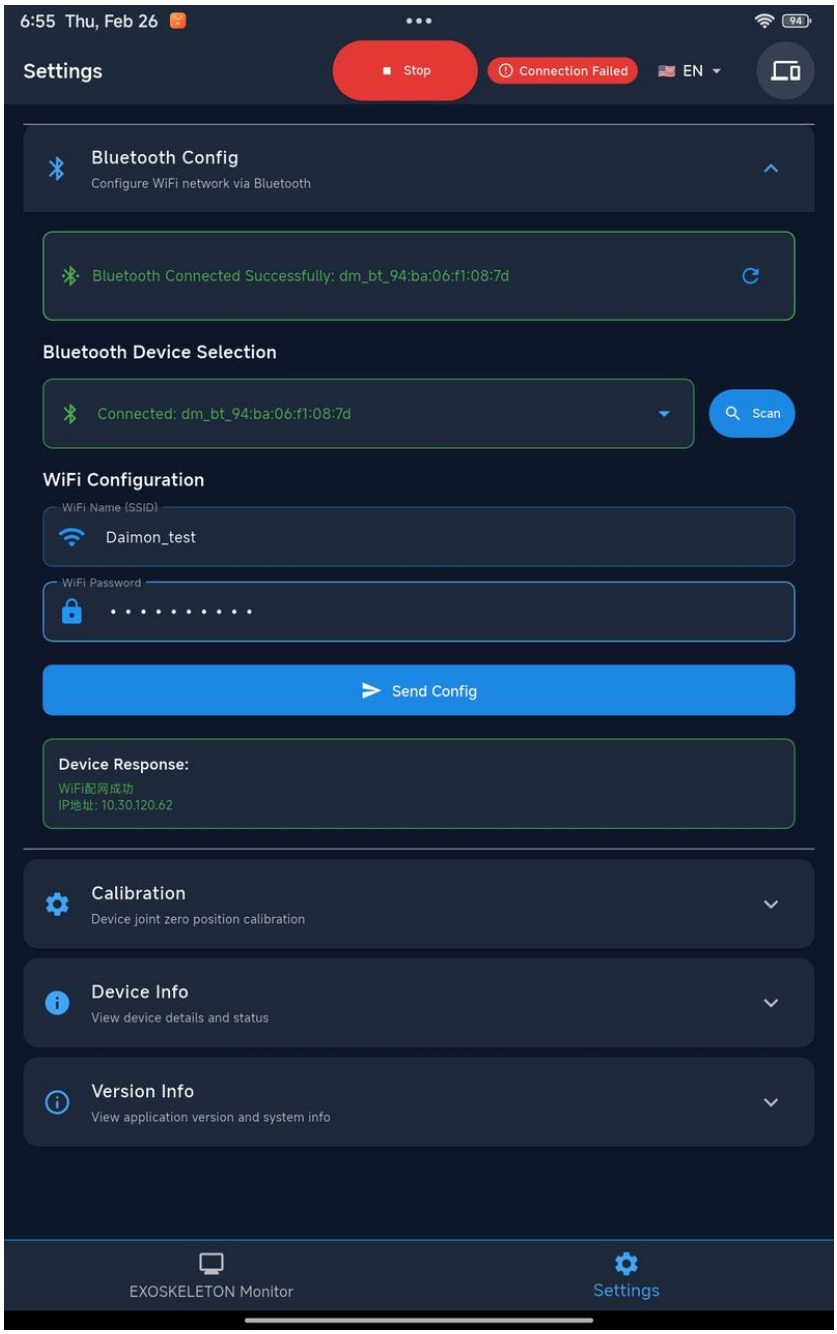
Click to scan for devices, select the Bluetooth of the previously scanned teleoperation data acquisition device, and connect.



Step 4:

Enter the ID and password of the target WIFI, and send the configuration. After a successful connection, the wireless IP obtained by the device will be returned.

If there is no returned IP after multiple network configuration attempts, please restart the remote operation data acquisition device and try again.



4.2 Software Installation and Deployment

Unzip DM-EXton2 Product Package/Install/install_ros2.zip or install_ros1.zip. It contains the following two components:

4.2.1 Message Bridge (MessageBridge)

Path: DM-EXton2 Product Package/Install/install_ros2/bin/ROS_2 or under install_ros1

Pre-compiled software packages are provided for Ubuntu 20 + ROS1 and Ubuntu 22 + ROS2.

Note: Bridge the remote operation data acquisition device with ROS messages.

Note: Python packages and source code are provided, including ZMQ metadata interface version, ROS1 bridge version, and ROS2 bridge version. It can be used on X86 or ARM platforms.

4.2.2 Control Plugin with Mirroring and Clutch Function (Puppetry)

路径： DM-EXton2 Product Package/Install/install_ros2/bin/Puppetry

Pre-compiled software packages are provided for Ubuntu 20 and Ubuntu 22.

Note: A more user-friendly incremental control teleoperation mode.

Note: For Ubuntu 22 on ARM architecture, pre-compiled software packages are available, which can be used after decompression.

4.3 Quick Start

Taking the Ubuntu 22 + ROS2 environment as an example:

4.3.1. Start the Message Bridge (MessageBridge)

Switch to the install_ros2/bin/ROS_2/ path:

Execute./MessageBridge (or Bridge.sh)

代码块

```
cd install_ros2/bin/ROS_2
./MessageBridge
```

Note: The appearance of the log MessageBridge Node 'message_bridge' in namespace '/' started in the last line of the log indicates success. (If started before self-check is completed, markers such as service registration fail will appear, and at this time the force feedback function cannot be used normally, only one-way teleoperation is available.)

代码块

```
./MessageBridge
[INFO] [1770704408.330312411] []: 命令行参数: monitoring_ip=127.0.0.1, advertise_ip=, bridge_port=9900, service_name=bridge_down_pub, wait_master_ms=0, show_raw=false
...
[INFO] [1770704408.753845198] []: ServiceRegistrar RegisterService: Service bridge_down_pub registered.
[INFO] [1770704408.753867633] []: Sent REGISTER request of service-bridge_down_pub done
[INFO] [1770704408.753910588] []: Service 'bridge_down_pub' registered to Hub at tcp://192.168.2.240:5555 (advertise_ip='192.168.2.100', port=9900)
[INFO] [1770704408.753916235] []: MessageBridge Node 'message_bridge' in namespace '/' started
```

4.3.2 Start the control plugin (Puppetry) with mirroring and clutch functions

Switch to the install_ros2/bin/Puppetry/ path:

Execute./Puppetry

```
代码块
1   cd install_ros2/bin/Puppetry/
2   ./Puppetry
```

Note: The appearance of the log "Puppetry initialized. Press Ctrl+C to stop." in the last line of the log indicates success.

```
代码块
1   ./Puppetry
2   ================================================================================
    ======
3
4   Application information:
5   Name: Puppetry
6   Version: 0.0.1
7    ...
8   [1770704441.135959][tid 3174399][I][Puppetry.Initialize 179] Puppetry
    initialized, task ID: 0, interval: 2ms
9   [1770704441.135960][tid 3174399][I][main 64] Puppetry initialized. Press Ctrl+C
     to stop.
```

4.4 Software Usage Instructions

4.4.1 Parameter Configuration

Message Bridge (MessageBridge)

```
代码块
1   参数文件位于：
2   Install/install_ros2/bin/ROS_2/config/MessageBridge.yaml
```

The parameter configuration file is as follows:

```
代码块
```

```yaml
# MessageBridge Configuration File

MessageBridge:
  # Network Configuration
  network:
    monitoring_ip: "192.168.2.240"    # IP address of the exoskeleton device
    advertise_ip: ""              # Local IP address to advertise (auto-detect if empty). For Docker, use host IP
    bridge_pub_port: 9900         # Port number for bridge publishing service

  # ROS Topic Name Configuration
  ros_topics:
    joints: "/joints"                           # Exoskeleton joint angles topic
    ee_poses: "/ee_poses"                       # Exoskeleton end-effector poses topic
    ee_trigger: "/trigger_positions"            # Trigger position data topic. First element: left hand, second: right hand. Range: 0-1
    joint_state: "/exoskeleton/joint_states"    # Exoskeleton joint states (TF) topic
    tact_force: "/tact_force"                   # Bilateral tactile force feedback topic. First element: left hand, second: right hand. Range: 0-1
    cmd_vel: "/custom/cmd_vel"                  # Velocity command topic
    cmd_vel2: "/custom/cmd_vel2"                # Velocity command topic 2
    data_record_enable: "/custom/data_record_enable"  # Data recording enable topic
    custom_a: "/custom/custom_a"               # Custom parameter A topic
    left_end_increment: "/custom/left_end_increment"   # Left end-effector incremental pose (raw data relative to previous frame)
    right_end_increment: "/custom/right_end_increment" # Right end-effector incremental pose (raw data relative to previous frame)
    clutch_left: "/clutch/left"                 # Left arm teleoperation enable signal (Clutch)
    clutch_right: "/clutch/right"               # Right arm teleoperation enable signal (Clutch)
    target_robot_left_ee_state: "/target_robot/left_ee/state"    # Target robot left end-effector pose state (subscribed)
    target_robot_right_ee_state: "/target_robot/right_ee/state"  # Target robot right end-effector pose state (subscribed)
    target_robot_left_ee_target: "/target_robot/left_ee/target"  # Target robot left end-effector pose target (published)
    target_robot_right_ee_target: "/target_robot/right_ee/target" # Target robot right end-effector pose target (published)

  # Frame ID Configuration
  frame_ids:
```

```
    # Base frame: Used for header.frame_id in incremental control pose messages
    # Affected topics: left_end_increment, right_end_increment, target_robot_left/right_ee_target
    # Must match the base frame name of the target robot
    base_frame: "base_link"

    # End-effector frames: Only used for TF broadcasting visualization, does not affect incremental control flow
    # TF tree structure: base_frame -> left_ee_frame / right_ee_frame
    left_ee_frame: "exo_left_ee_frame"    # Left end-effector frame name (for TF visualization)
    right_ee_frame: "exo_right_ee_frame"  # Right end-effector frame name (for TF visualization)

```

Key Parameter Settings:

monitoring_ip: It needs to be set to the IP Address of the remote operation data acquisition device.

advertise_ip: Automatically detected by the program, usually can be left blank. When running in a Docker environment without network mounting, it needs to be manually passed in.

base_frame: If using the incremental control plugin, the reference coordinate system for forward/inverse kinematics of the left and right hands needs to be filled in.

ros_topics: Modify the topic name according to requirements.

(Same Python version)

If you do not wish to modify relevant parameters via a parameter file, you can pass the following parameters when running the command:

代码块

```
./MessageBridge -h
Usage: MessageBridge [--help] [--version] [--monitoring-ip VAR] [--advertise-ip VAR] [--bridge-port VAR] [--service-name VAR] [--wait-master-ms VAR] [-show_raw]

Optional arguments:
  -h, --help        shows help message and exits
  -v, --version     prints version information and exits
  --monitoring-ip   [nargs=0..1] [default: "127.0.0.1"]
  --advertise-ip    [nargs=0..1] [default: ""]
  --bridge-port     [nargs=0..1] [default: 9900]
```

```
    --service-name    [nargs=0..1] [default: "bridge_down_pub"]
    --wait-master-ms  [nargs=0..1] [default: 0]
    -show_raw
```

Puppetry Incremental Control Plugin

代码块

```
参数文件位于:
Install/install_ros2/bin/Puppetry/config/Puppetry.yaml
```

The parameter configuration file is as follows:

代码块

```yaml
Puppetry:
  communication:
    puppeteer_monitoring_ip: "192.168.2.240"
    puppeteer_port: 9899
    use_ipc: true
    messagebridge_ipc_path: "ipc:///tmp/messagebridge_pub.ipc"
    puppetry_ipc_path: "ipc:///tmp/puppetry_pub.ipc"
    messagebridge_tcp_ip: "127.0.0.1"
    messagebridge_port: 9900
  urdf:
    directory: "./urdf_models"
    default_file: "robot.urdf"
  kinematics:
    enable_ik_solver: true
    solver_type: "pinocchio"
    max_iterations: 1000
    tolerance: 1.0e-6
    use_joint_limits: true
    collision_check: false
  logging:
    level: "info"  # trace, debug, info, warn, error, critical
  mirroring:
    enable_mirroring: false   # Mirroring disabled by default
  run:
    # run interval (ms). Original exoskeleton feedback is about 1000Hz (1ms)
    # Users can configure within the range of 1~1000ms (1kHz ~ 1Hz)
    interval_ms: 2

```

The parameters to be configured are as follows:

puppeteer_monitoring_ip: It needs to be configured as the IP of the remote operation data collection device.

enable_mirroring: Whether to enable the mirroring control mode.

interval_ms: Controls the running frequency and affects the frequency of the incremental control topic.

4.4.2 ROS API

Topic (Preset Name of Configuration File)	msg format	Transmission and reception type	Description
/joints	Float64MultiArray	Send	Remote operation of data acquisition device joint angle, 12-bit
/ee_poses	Float32MultiArray	Send	Remote operation data acquisition device left and right hand end pose (t.x y z r.w x y z), 14 bits
TF	TF	Send	TF publication of the end pose of the left and right hands of the teleoperated data acquisition device, same as the previous data
/trigger_positions	Float64MultiArray	Send	Remote operation data acquisition device trigger position, range 0 - 1.0, 2 digits
/exoskeleton/joint_states	JointState	Send	Joint state of the remotely operated data acquisition device
/tact_force	Float64MultiArray	Received	Force feedback of the left and right hand tactile sensors of the robot, range 0-1.0, 2 digits
/custom/cmd_vel	Twist	Send	Speed interface controlled by the cross key on the left joystick of the teleoperation data acquisition device, Linear.x + Angular.z
/custom/cmd_vel2	Twist	Send	Speed interface controlled by the cross key on the right joystick of the teleoperation data acquisition device, Linear.x + Linear.y
	Bool		

/custom/data_record_enable		Send	Data recording start/stop topic for remote operation of data acquisition device handle control
/custom/left_end_increment	PoseStamped	Send	Metadata of the increment of the left-hand end pose of the teleoperated data acquisition device relative to the previous frame of data
/custom/right_end_increment	PoseStamped	Send	Metadata of the increment of the pose of the right-hand end of the remotely operated data acquisition device relative to the previous frame of data
/clutch/left	Bool	Send	Signal of the left-hand enable side key press on the teleoperation data acquisition device
/clutch/right	Bool	Send	Signal of the left-hand enable side key press on the teleoperation data acquisition device
/target_robot/left_ee/state	PoseStamped	Received	Current pose of the left hand end of the target robot
/target_robot/right_ee/state	PoseStamped	Received	Current pose of the end of the right hand of the target robot
/target_robot/left_ee/target	PoseStamped	Send	Target pose of the left end of the target robot (messages will appear if the incremental plugin is activated)
/target_robot/right_ee/target	PoseStamped	Send	Target pose of the right end of the target robot (messages will appear if the incremental plugin is activated)

4.4.3 Python Version Message Bridge

The Python version bridge starts as follows (taking the ROS2 version as an example):

解压DM-EXton2 Product Package/install_py/message-bridge-py-develop_ros2.zip
使用方法：

代码块

```
cd install_py/message-bridge-py-develop_ros2/

source /opt/ros/humble/setup.bash

pip3 install pyzmq pyyaml

```

```
pip3 install .

python3 -m message_bridge
```

Other versions are as follows:

代码块

```
# Basic Version (develop_base) - No ROS required:

# Install dependencies
pip3 install pyzmq pyyaml
# Install the project
pip3 install .


# ROS1 Version (develop_ros1) - ROS Noetic required:
# Set up the ROS environment first
source /opt/ros/noetic/setup.bash
# Install dependencies
pip3 install pyzmq pyyaml
# Install the project
pip3 install .


# ROS2 Version (develop_ros2) - ROS2 Humble/Iron required:
# Set up the ROS2 environment first
source /opt/ros/humble/setup.bash
# Install dependencies
pip3 install pyzmq pyyaml
# Install the project
pip3 install .


# Basic operation
python3 -m message_bridge
# Specify IP (override configuration file)
python3 -m message_bridge --ip 10.30.120.101
# Debug mode (display raw data)
python3 -m message_bridge --ip 10.30.120.101 --show-raw
# Disable ServiceHub registration
python3 -m message_bridge --ip 10.30.120.101 --no-service-hub
```

4.5 Encoder Zero Calibration

Zero calibration has been completed at the factory. Under normal circumstances, it is generally unnecessary to perform zero calibration again, and it is only carried out when the joints of the equipment have been disassembled and reassembled during operation and maintenance.

Power On: Hang or lay flat the backpack base;

Remove the cylindrical pin from the plastic bag and secure the shoulder joint as shown in the figure.

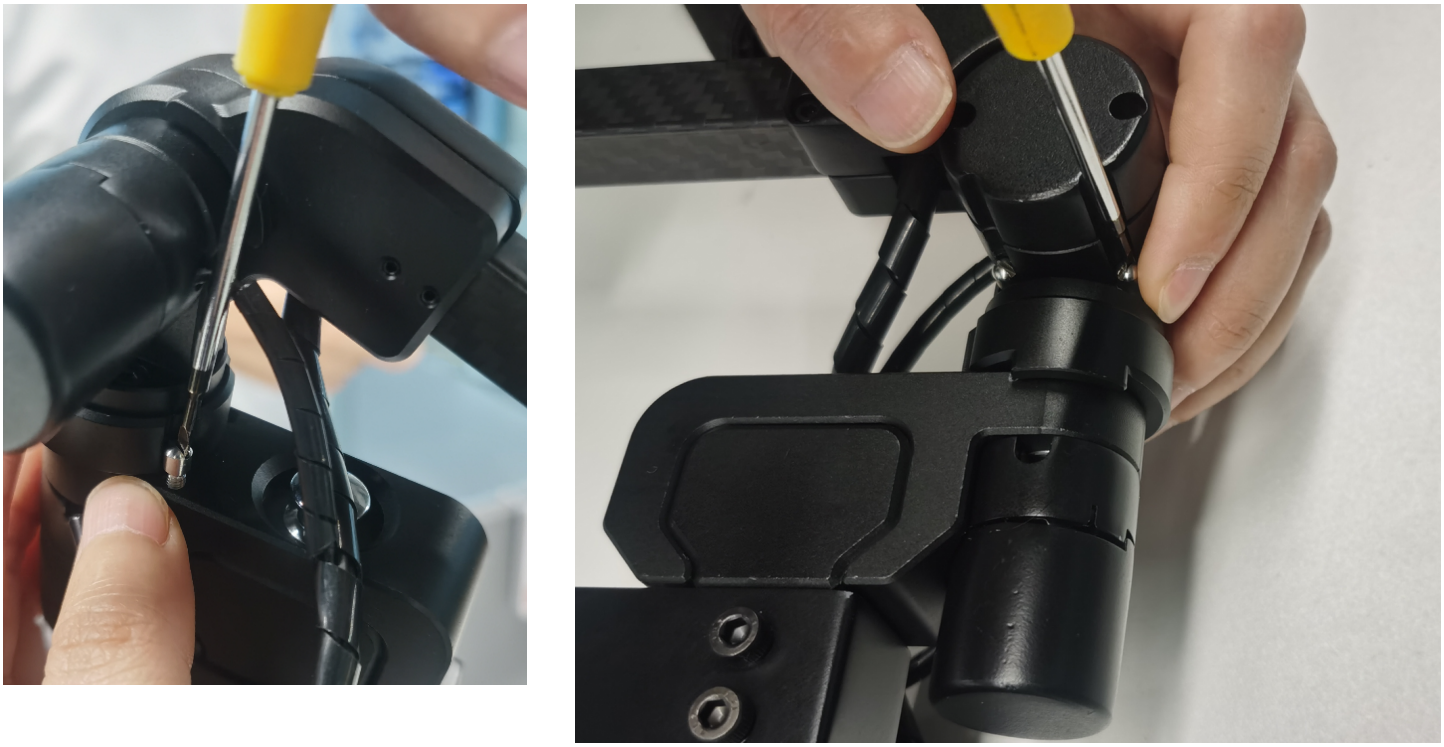


Figure 4-2 Cylindrical Pin Fixing Shoulder Joint

Method for removing and locking the zero-position tool: Assemble it by referring to the figure below.



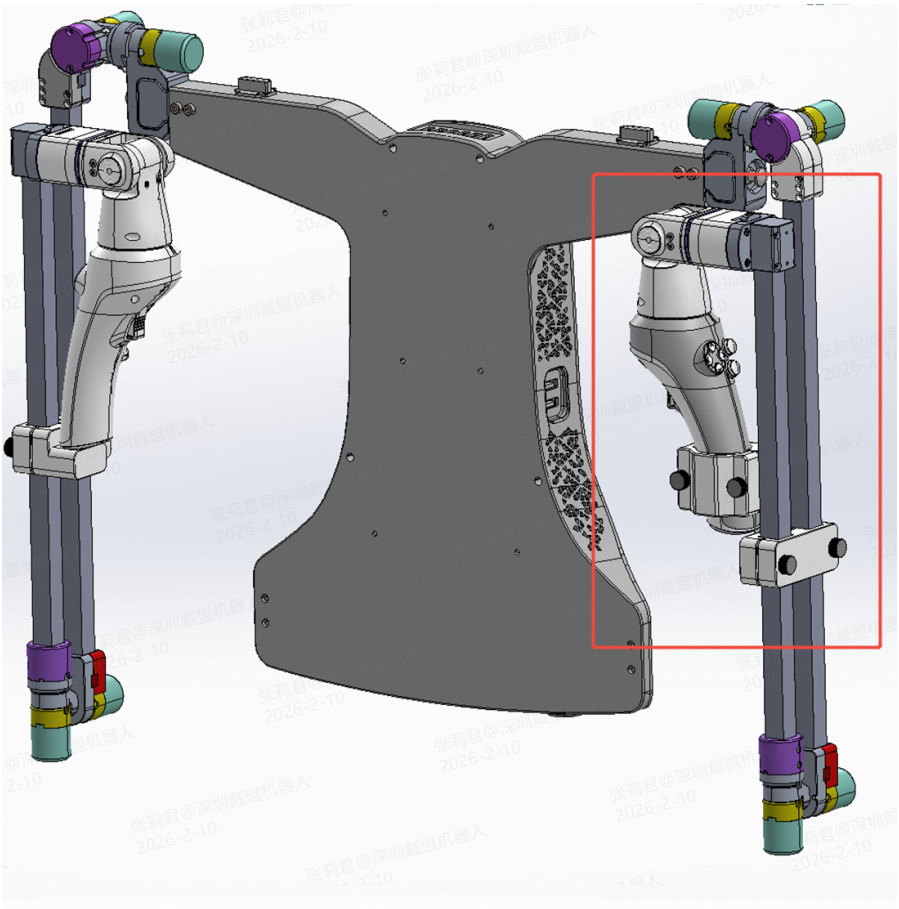




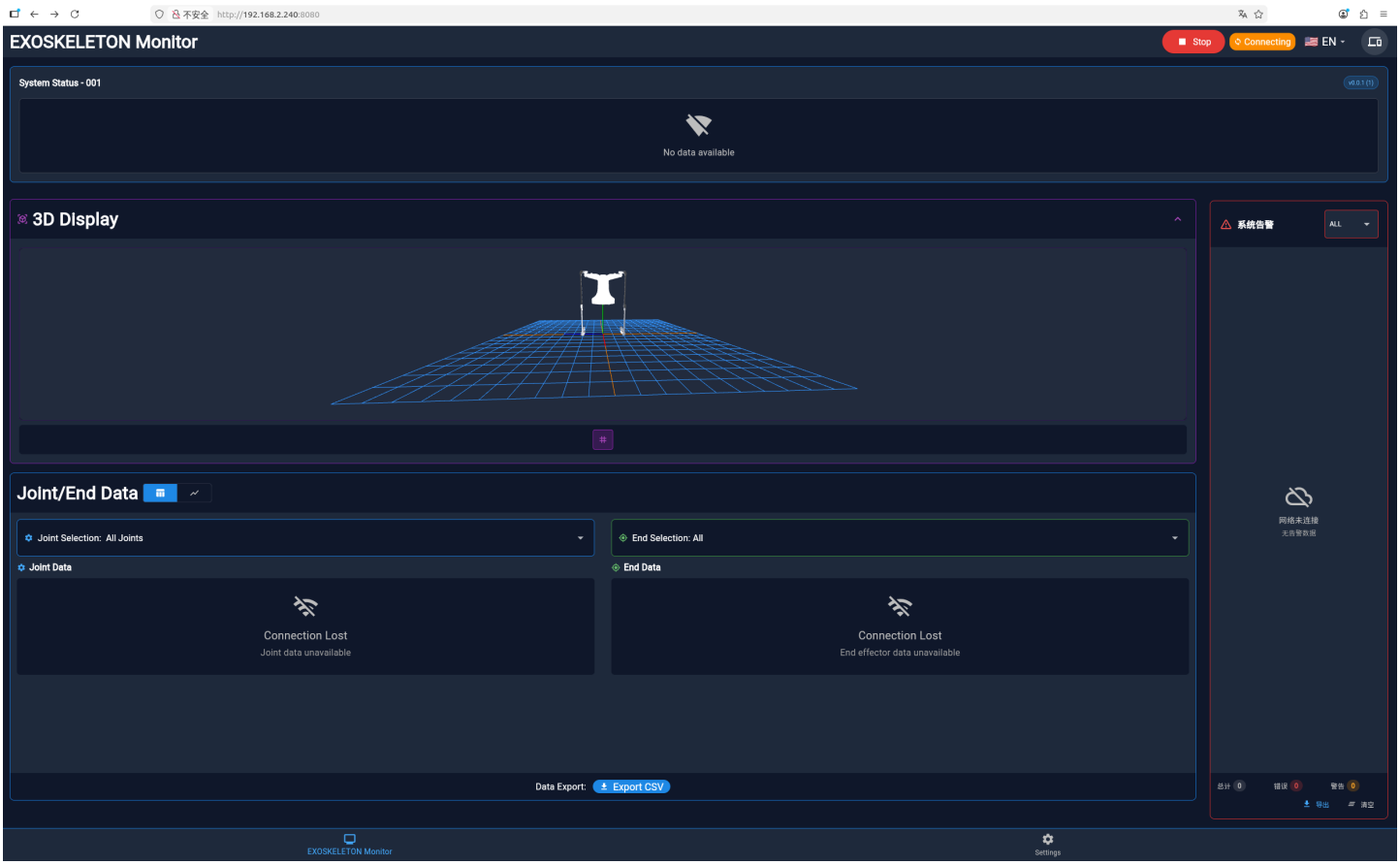
Figure 4-3 Installation Diagram of Zero Calibration Tool

After the device has completed zero calibration installation, log in to the WEB page settings menu item to perform zero setting.

Open a browser (Chrome is recommended), and enter the corresponding IP address and port:

```
代码块
1   192.168.2.240:8080
```

The opened page is as follows:



Click the device management button in the upper right corner:

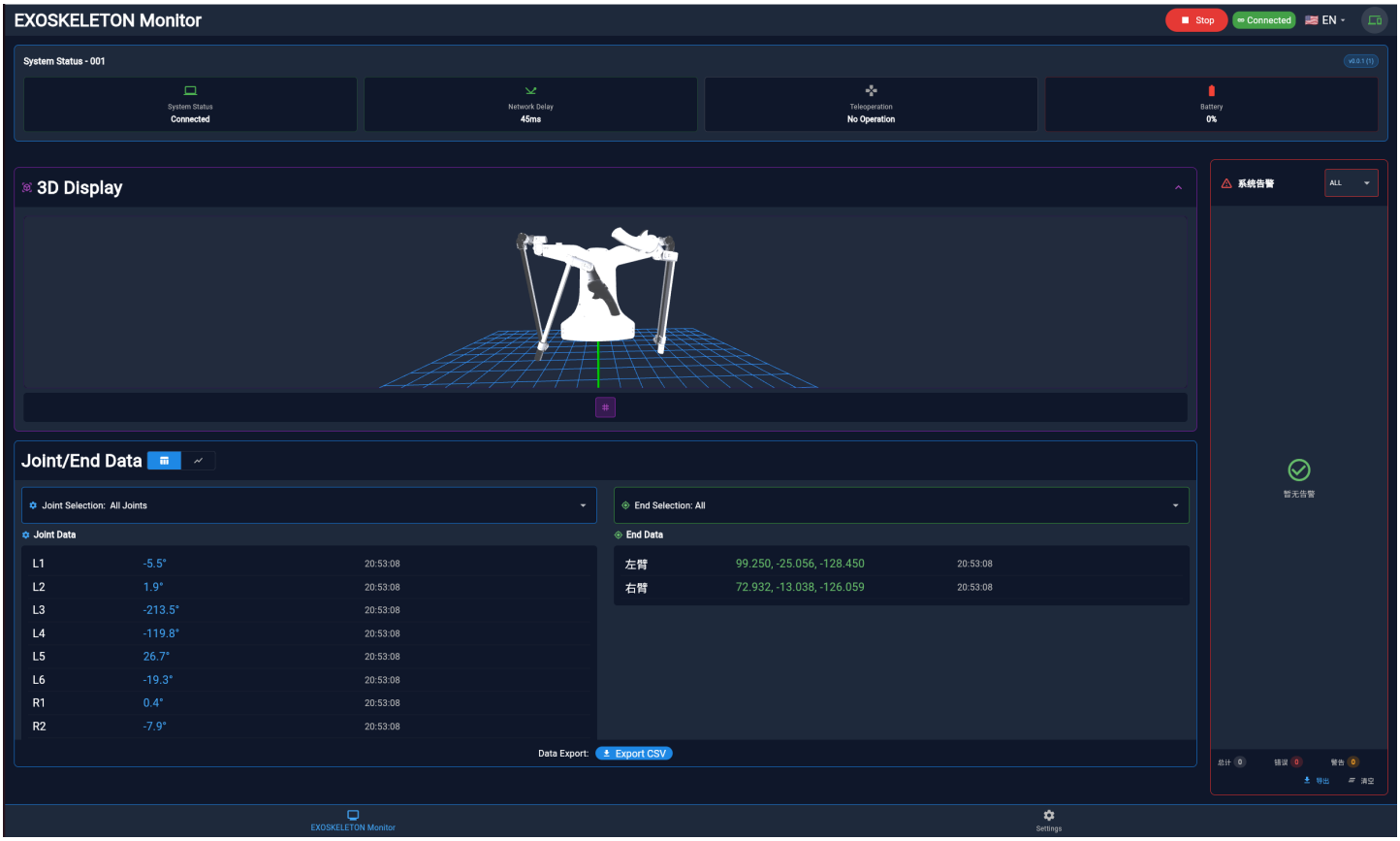


Enter the corresponding IP and connect:

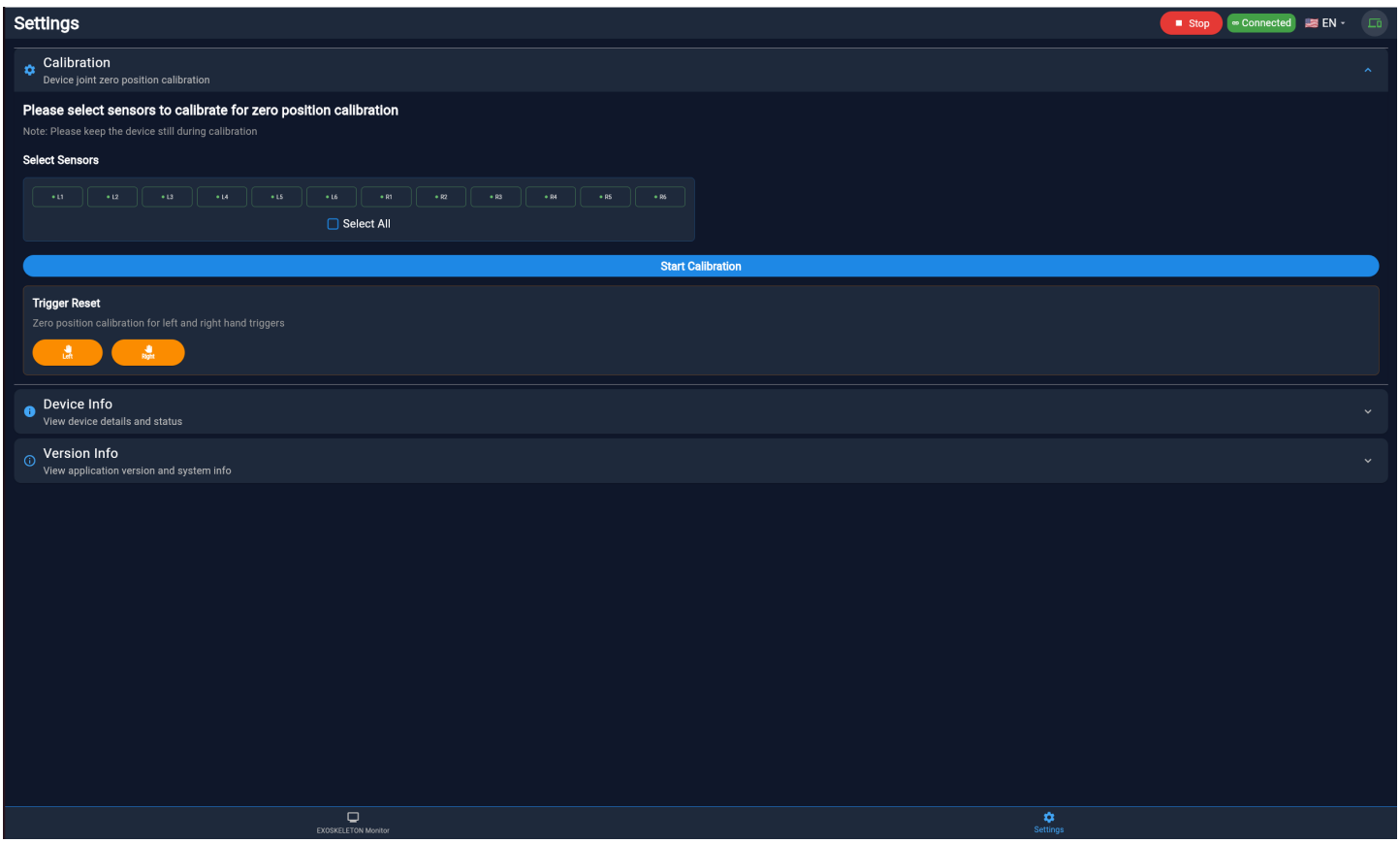
(Port is fixed at 9002)



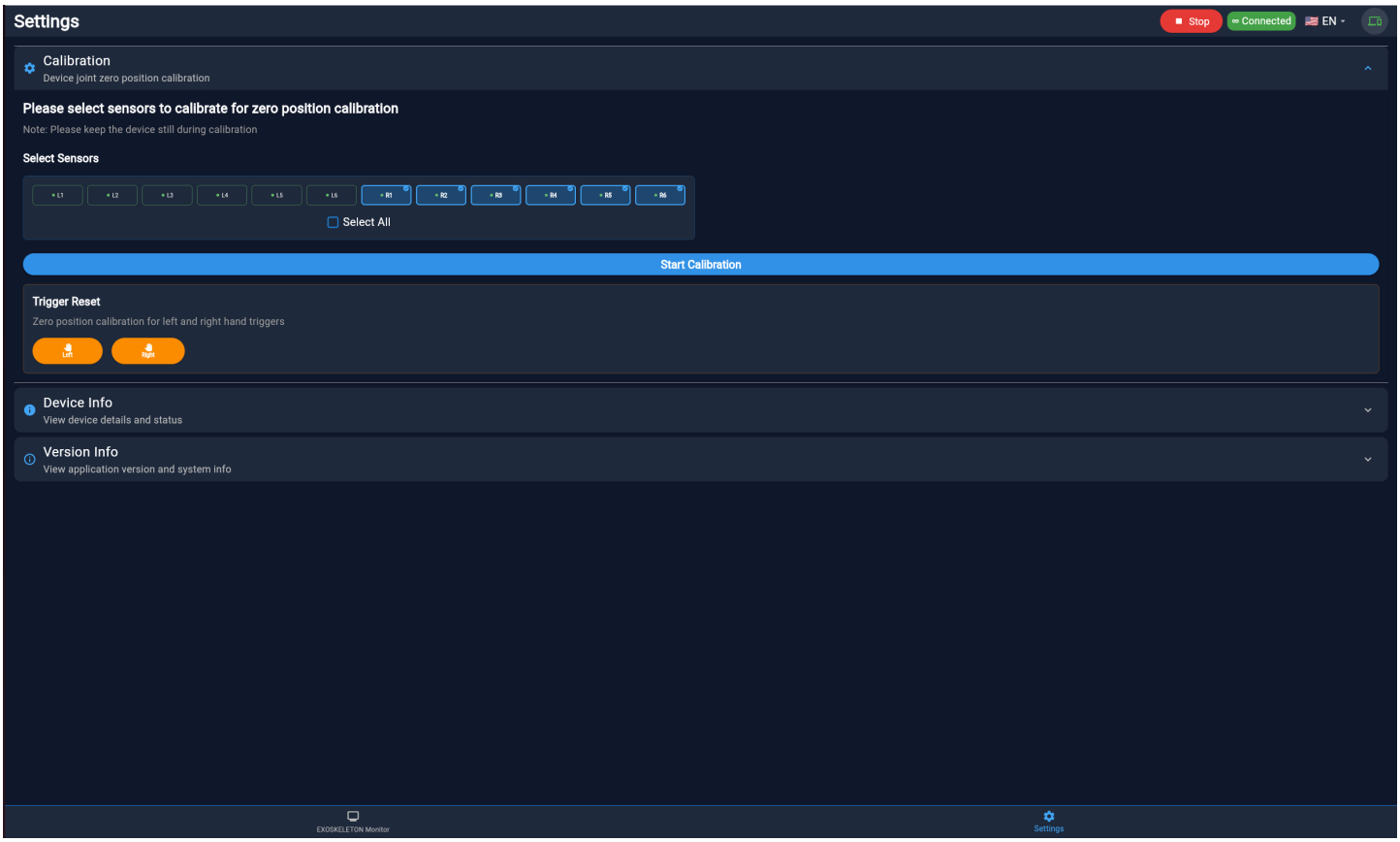
After successful connection, the page will display as follows:



Switch to the settings page below and turn on calibration at position 0:



Select the joint you want to recalibrate and click "Start Calibration":



After clicking "Start Calibration", the calibration fixture can be removed.

Generally speaking, the calibration results can be observed through RVIZ. The web page model will be updated after refreshing the page once calibration is complete.

5 Safety and Precautions

5.1 Precautions for Hardware Use

1. Please follow the requirements of the joint angle range shown in 4.1.1 during use to avoid damaging the equipment.
2. Ensure that each wire is not directly stressed to prevent wire breakage.

5.2 Safety Warning

Prevent children from misusing

This product contains small parts, which pose a risk of accidental ingestion. Please store this product and its accessories in a place out of reach of children to prevent accidental ingestion or misuse. If a part is accidentally ingested, seek medical attention immediately.

6 Maintenance and Upkeep

6.1 Routine Maintenance

1. Regularly clean the surface of the remote operation data acquisition equipment, wipe it with a clean damp cloth, avoid using corrosive cleaners, and avoid direct flushing or soaking for cleaning.
2. Before use, check the fixing straps and joint components. If any damage is found, replace them promptly.
3. Regularly check the fixing screws, and replace them with new ones promptly if they are loose.
4. When the joint rotation axis is stuck, mechanical equipment lubricating oil can be used for lubrication.

6.2 Battery Management

1. Please use the charger originally configured by Daimon for charging, and avoid using third-party chargers of unknown specifications to prevent damage to the built-in battery.
2. It is recommended that the charging environment of the lithium battery pack be maintained at 10°C - 35°C, and long-term charging after full charge should be avoided.
3. If a charging anomaly is detected, please immediately stop using it and contact after-sales service.
4. Do not power on and use the remote control device while charging to prevent the battery from overheating and malfunctioning.

6.3 Storage Specification

1. Environmental Requirements: Temperature -20°C to 50°C, Humidity ⩽70%RH, Keep Away from Strong Magnetic Fields.
2. Avoid impact and vibration during transportation to prevent damage.

7 Disclaimer

1. **Responsibility for Safe Use:**

Users shall use and operate this device in a safe environment and strictly comply with all operating specifications in the instruction manual. This system is applicable to specified uses such as remote control of robots, Data Acquisition, and learning and training, and is strictly prohibited from being used in hazardous operations or other unauthorized scenarios. If a user operates beyond the safe environment without authorization, the user shall bear full responsibility for all damages caused thereby.

2. **Third Party Liability Exclusion:**

The Company shall not assume any direct or indirect liability for personal injury, property damage, data loss, or other consequences caused by the following circumstances:

- Improper operation by the user: including but not limited to failure to perform equipment maintenance as required by the instruction manual, failure to regularly calibrate system parameters, unauthorized modification of system configuration, etc.;
- Force Majeure and Environmental Factors: Force majeure events such as natural disasters, wars, government regulations, electromagnetic interference, extreme weather, etc., or the operating environment failing to meet the requirements of this system (such as high temperature, humidity, strong magnetic field environment, etc.);
- Third - party product issues: Malfunctions and damages caused by compatibility issues between the software, hardware devices, and network services provided by third - parties and this device, or by users' unauthorized use of non - officially certified accessories or non - authorized expansion modules;
- Accidents and human factors: including but not limited to damage caused by accidental events or human factors such as device dropping, collision, water immersion, theft, malicious attacks, etc.

3. **Warranty and Scope of Liability:**

- This device is covered by a 12-month limited warranty under normal usage conditions. During the warranty period, users can only enjoy the warranty service by presenting a valid purchase receipt, provided that the device has not malfunctioned due to human damage, unauthorized disassembly, improper operation, or other such circumstances.
- Users must strictly follow the instructions for operation. Modifying, disassembling the equipment on their own or using non-original accessories is prohibited. If a safety accident, equipment damage, or data loss occurs due to violation of the operating specifications, the user shall bear full responsibility, and the equipment will automatically lose its warranty eligibility.
- Damage beyond the warranty scope, problems caused by user operation errors, and consequences arising from repairs not carried out through the officially designated after-sales channels are not within the scope of after-sales service responsibilities.

4. **Product Performance:**

The Company is committed to ensuring the accuracy and stability of this product. All performance parameters of the product are measured under laboratory test conditions. The Company shall not be liable for any non-compliance with functionality or performance that may occur in special application scenarios. Users are required to independently evaluate whether this product meets their actual application requirements.

5. **Intellectual Property:**

The design, technology, and all related materials and rights of this product are owned by our company. Any form of reverse engineering, technical decryption, and unauthorized commercial use are prohibited.

6. **Limitation of Liability:**

Unless otherwise provided by law, the Company shall not be liable for any indirect losses, special damages or incidental damages, including but not limited to loss of profits, data loss or business interruption, etc.

Final right of interpretation:

The final interpretation right of the disclaimer shall be reserved by Daimon (Shenzhen) Robotics Technology Co., Ltd. within the scope permitted by law. The Company reserves the right to revise and update this statement based on actual circumstances, and the revised content will be announced through the official website or other appropriate channels. Continued use of this system by users shall be deemed as acceptance of the revised terms.

8 Contact Information

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